CSH3143 - Knowledge Representation and Programming in Logic

Knowledge Representation: Facts and Rules - Practical Exercises

Instructions: Represent each problem using suitable Prolog facts and rules. Test your program using appropriate queries.

Exercise 1 - Family Relationship

Consider the following family information:

- Ahmed is the parent of Fathima.
- Ahmed is the parent of Nimal.
- Fathima is the parent of Sara.
- Nimal is the parent of Kamal.

Represent the information using suitable Prolog facts and write rules to determine:

1. Whether one person is the **child** of another person.
2. Whether two people are **siblings**.
3. Whether one person is the **grandparent** of another person.

Test your program using suitable queries.

Exercise 2 - University Course Eligibility

A university has the following rules:

- A student can register for **Artificial Intelligence** if the student has passed Programming and Mathematics.
- A student can register for **Machine Learning** if the student has passed Artificial Intelligence and Statistics.

The following information is known:

- Amal has passed Programming.
- Amal has passed Mathematics.
- Amal has passed Statistics.
- Sara has passed Programming.
- Sara has passed Mathematics.
- Nimal has passed Programming.

Represent the knowledge using Prolog **facts and rules**.

Write rules to determine whether a student is eligible for:

```
eligible_ai(Student)
eligible_ml(Student)
```

Use suitable queries to identify all eligible students.

Exercise 3 - Animal Classification

Consider the following knowledge:

- A parrot has feathers and can fly.
- An eagle has feathers and can fly.
- A penguin has feathers but cannot fly.
- A tiger has hair and gives birth to young.
- A dog has hair and gives birth to young.

Use Prolog facts and rules to represent the knowledge.

Define rules such that:

- An animal with feathers is classified as a **bird**.
- An animal with hair and which gives birth to young is classified as a **mammal**.
- A bird that can fly is classified as a **flying bird**.

Your program should be able to answer queries such as:

```
?- bird(parrot).
?- mammal(tiger).
?- flying_bird(X).
```

Exercise 4 - Library Membership

A university library maintains the following information:

- Amal is a student.
- Sara is a student.
- Kamal is a lecturer.
- Nimal is a visitor.
- Amal has no outstanding library fine.
- Sara has an outstanding fine.
- Kamal has no outstanding fine.

The library follows these rules:

- Students and lecturers are library members.
- A library member can borrow a book only if there is **no outstanding fine**.
- Visitors cannot borrow books.

Represent the above knowledge using facts and rules.

Write the following rules:

```
library_member(Person)
can_borrow(Person)
```

Test your program to determine who can borrow books.

Exercise 5 - Traffic Control Rules

Consider a simple traffic control system.

The following conditions may occur:

```
Traffic light is red
Traffic light is yellow
Traffic light is green
Pedestrian signal is active
```

Represent these conditions using appropriate facts.

Create rules with the following knowledge:

- If the traffic light is red, vehicles must stop.
- If the traffic light is green, vehicles may proceed.
- If the traffic light is yellow, vehicles must slow down.
- If the pedestrian signal is active, vehicles must stop.

Write a rule:

```
vehicle_action(Action)
```

that determines the appropriate action.

Use suitable queries to test your knowledge base.

Exercise 6 - Scholarship Selection

The following information is available about students:

Student	GPA	Attendance	Income Level
amal	3.8	90	low
sara	3.2	88	low
nimal	3.7	75	high
kamal	3.9	92	low

Represent this information using Prolog facts.

Use the following rules:

- A student has **good academic performance** if the GPA is 3.5 or above.
- A student has **good attendance** if attendance is 80% or above.
- A student is **scholarship eligible** if the student has good academic performance and good attendance.
- A student receives **priority scholarship consideration** if the student is scholarship eligible and belongs to the low-income category.

Write appropriate Prolog rules and determine:

```
?- scholarship_eligible(X).
?- priority_scholarship(X).
```

Exercise 7 - Employee Promotion

A company stores information about its employees:

```
amal - 6 years experience - performance 85
sara - 3 years experience - performance 90
nimal - 8 years experience - performance 70
kamal - 7 years experience - performance 92
```

Represent these details using Prolog facts.

The company has the following rules:

- An employee is **experienced** if the employee has at least 5 years of experience.
- An employee has **good performance** if the performance mark is 80 or above.
- An employee is **eligible for promotion** if the employee is experienced and has good performance.

Create suitable rules and determine all employees who are eligible for promotion.

Exercise 8 - Map Colouring

Consider five regions: **A, B, C, D and E**.

The neighbouring relationships are:

- A is adjacent to B and C.
- B is adjacent to A, C and D.
- C is adjacent to A, B and E.
- D is adjacent to B and E.
- E is adjacent to C and D.

The available colours are:

```
red
green
blue
```

Represent:

1. The available colours as facts.

2. The neighbouring relationships as facts.

3. A rule stating that adjacent regions cannot have the same colour.

4. A rule `colour_map(A, B, C, D, E)` which finds a valid colour assignment for all five regions.

Test the program using:

```
?- colour_map(A, B, C, D, E).
```